

Time-Independent Schrodinger's Equation

Quantum Mechanics

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1 Stationary States

For time-independent potentials $V(x)$, the Schrödinger equation can be solved using **separation of variables**. We seek solutions of the form:

$$\Psi(x, t) = \psi(x)f(t). \quad (1)$$

Substituting into the time-dependent Schrödinger equation and dividing by ψf yields:

$$i\hbar \frac{1}{f} \frac{df}{dt} = -\frac{\hbar^2}{2m} \frac{1}{\psi} \frac{d^2\psi}{dx^2} + V(x). \quad (2)$$

The left side depends only on t , the right side only on x . Hence both sides must equal a constant, called the **separation constant** E :

$$i\hbar \frac{df}{dt} = Ef, \quad (3)$$

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V\psi = E\psi. \quad (4)$$

The time equation gives:

$$f(t) = e^{-iEt/\hbar}. \quad (5)$$

The spatial equation is the **time-independent Schrödinger equation**:

$$\boxed{-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V\psi = E\psi}. \quad (6)$$

1.1 Properties of Separable Solutions

1. **Stationary states:** The probability density is time-independent:

$$|\Psi(x, t)|^2 = |\psi(x)|^2.$$

All expectation values are constant in time.

2. **Definite energy:** The Hamiltonian operator $\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x)$ satisfies $\hat{H}\psi = E\psi$. The variance of H is zero, so every measurement of total energy yields E .
3. **General solution:** The most general solution is a linear combination:

$$\Psi(x, t) = \sum_{n=1}^{\infty} c_n \psi_n(x) e^{-iE_n t/\hbar}. \quad (7)$$

The coefficients c_n are determined by the initial condition $\Psi(x, 0) = \sum c_n \psi_n(x)$.

1.2 Important Theorems

- For normalizable solutions, E must be real.
- $\psi(x)$ can always be taken as real (for non-degenerate states).
- If $V(x)$ is even, $\psi(x)$ can be chosen as even or odd.
- $E > V_{\min}$ for all normalizable solutions.

2 Infinite Square Well

The potential is:

$$V(x) = \begin{cases} 0, & 0 \leq x \leq a, \\ \infty, & \text{otherwise.} \end{cases}$$

Inside the well, the TISE becomes:

$$\frac{d^2\psi}{dx^2} = -k^2\psi, \quad k = \frac{\sqrt{2mE}}{\hbar}.$$

General solution: $\psi(x) = A \sin(kx) + B \cos(kx)$. Boundary conditions $\psi(0) = \psi(a) = 0$ give:

$$B = 0, \quad \sin(ka) = 0 \Rightarrow k_n = \frac{n\pi}{a}, \quad n = 1, 2, 3, \dots$$

Quantized energies:

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}. \quad (8)$$

Normalized stationary states:

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right). \quad (9)$$

2.1 Orthonormality and Completeness

The wave functions satisfy:

$$\int_0^a \psi_m^*(x) \psi_n(x) dx = \delta_{mn}. \quad (10)$$

Any function $f(x)$ on $[0, a]$ can be expanded:

$$f(x) = \sum_{n=1}^{\infty} c_n \psi_n(x), \quad c_n = \int_0^a \psi_n^*(x) f(x) dx.$$

3 The Harmonic Oscillator

Potential: $V(x) = \frac{1}{2}m\omega^2 x^2$, with $\omega = \sqrt{k/m}$. The TISE is:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2 x^2 \psi = E\psi. \quad (11)$$

3.1 Algebraic Method (Ladder Operators)

Define raising and lowering operators:

$$a_{\pm} \equiv \frac{1}{\sqrt{2m}} \left(\frac{\hbar}{i} \frac{d}{dx} \pm im\omega x \right). \quad (12)$$

They satisfy:

$$a_- a_+ = \frac{1}{2m} \left[\left(\frac{\hbar}{i} \frac{d}{dx} \right)^2 + (m\omega x)^2 \right] + \frac{1}{2} \hbar \omega. \quad (13)$$

The Hamiltonian can be written:

$$\hat{H} = \hbar\omega \left(a_+ a_- + \frac{1}{2} \right) = \hbar\omega \left(a_- a_+ - \frac{1}{2} \right). \quad (14)$$

The commutation relation is:

$$[a_-, a_+] = \hbar\omega. \quad (15)$$

If ψ has energy E , then $a_+\psi$ has energy $E + \hbar\omega$ (raising), and $a_-\psi$ has energy $E - \hbar\omega$ (lowering).

The ground state satisfies $a_-\psi_0 = 0$:

$$\frac{d\psi_0}{dx} = -\frac{m\omega}{\hbar} x \psi_0 \Rightarrow \psi_0(x) = A_0 e^{-m\omega x^2/(2\hbar)}.$$

Ground state energy:

$$E_0 = \frac{1}{2} \hbar\omega. \quad (16)$$

Excited states:

$$\psi_n(x) = A_n (a_+)^n e^{-m\omega x^2/(2\hbar)}, \quad E_n = \left(n + \frac{1}{2} \right) \hbar\omega. \quad (17)$$

The ladder operator relations:

$$a_+ \psi_n = i\sqrt{(n+1)\hbar\omega} \psi_{n+1}, \quad (18)$$

$$a_- \psi_n = -i\sqrt{n\hbar\omega} \psi_{n-1}. \quad (19)$$

3.2 Analytic Method (Hermite Polynomials)

Introduce dimensionless variable $\xi = \sqrt{m\omega/\hbar} x$. The TISE becomes:

$$\frac{d^2\psi}{d\xi^2} = (\xi^2 - K)\psi, \quad K = \frac{2E}{\hbar\omega}.$$

Substituting $\psi(\xi) = h(\xi)e^{-\xi^2/2}$ gives Hermite's equation:

$$\frac{d^2h}{d\xi^2} - 2\xi \frac{dh}{d\xi} + (K - 1)h = 0.$$

The power series solution yields quantization: $K = 2n + 1$, so $E_n = (n + \frac{1}{2})\hbar\omega$. The solutions are Hermite polynomials $H_n(\xi)$. The normalized wave functions:

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n(\xi) e^{-\xi^2/2}. \quad (20)$$

First few Hermite polynomials:

$$H_0(\xi) = 1, \quad H_1(\xi) = 2\xi, \quad H_2(\xi) = 4\xi^2 - 2, \quad H_3(\xi) = 8\xi^3 - 12\xi.$$

4 Free Particle

For $V(x) = 0$, the TISE is:

$$\frac{d^2\psi}{dx^2} = -k^2\psi, \quad k = \frac{\sqrt{2mE}}{\hbar}.$$

General solution: $\psi(x) = Ae^{ikx} + Be^{-ikx}$. Stationary states:

$$\Psi_k(x, t) = Ae^{i(kx - \frac{\hbar k^2}{2m}t)}. \quad (21)$$

These are **not normalizable** (plane waves). The general solution is a **wave packet**:

$$\Psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(k) e^{i(kx - \frac{\hbar k^2}{2m}t)} dk. \quad (22)$$

Given $\Psi(x, 0)$, the Fourier transform gives:

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \Psi(x, 0) e^{-ikx} dx.$$

4.1 Phase and Group Velocities

For a wave packet, peaked at k_0 :

- **Phase velocity:** $v_{\text{phase}} = \frac{\omega}{k} = \frac{\hbar k}{2m}$.
- **Group velocity:** $v_{\text{group}} = \frac{d\omega}{dk} = \frac{\hbar k}{m}$.

The group velocity matches the classical particle speed: $v_{\text{classical}} = \sqrt{2E/m} = \hbar k/m$.

5 Delta-Function Potential

Consider $V(x) = -\alpha\delta(x)$ (attractive well), with $\alpha > 0$.

5.1 Bound States ($E < 0$)

For $x \neq 0$, the TISE gives $\psi'' = \kappa^2\psi$ with $\kappa = \sqrt{-2mE}/\hbar$. The normalizable solution is:

$$\psi(x) = \begin{cases} Be^{\kappa x}, & x < 0, \\ Be^{-\kappa x}, & x > 0. \end{cases}$$

Integrating the TISE across $x = 0$ gives the discontinuity condition:

$$\Delta \left(\frac{d\psi}{dx} \right) = -\frac{2m\alpha}{\hbar^2} \psi(0). \quad (23)$$

Applying this yields $\kappa = m\alpha/\hbar^2$, so:

$$E = -\frac{m\alpha^2}{2\hbar^2}, \quad \psi(x) = \frac{\sqrt{m\alpha}}{\hbar} e^{-m\alpha|x|/\hbar^2}. \quad (24)$$

Exactly one bound state exists regardless of α .

5.2 Scattering States ($E > 0$)

For $E > 0$, $k = \sqrt{2mE}/\hbar$. The wave function:

$$\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx}, & x < 0, \\ Fe^{ikx} + Ge^{-ikx}, & x > 0. \end{cases}$$

For scattering from the left ($G = 0$), continuity and the jump condition give:

$$B = \frac{i\beta}{1 - i\beta}A, \quad F = \frac{1}{1 - i\beta}A, \quad \beta = \frac{m\alpha}{\hbar^2 k}.$$

Reflection and transmission coefficients:

$$R = \frac{|B|^2}{|A|^2} = \frac{\beta^2}{1 + \beta^2} = \frac{1}{1 + \frac{2\hbar^2 E}{m\alpha^2}}, \quad T = \frac{|F|^2}{|A|^2} = \frac{1}{1 + \beta^2} = \frac{1}{1 + \frac{m\alpha^2}{2\hbar^2 E}}. \quad (25)$$

Note: $R + T = 1$. For a delta-function barrier ($V = +\alpha\delta(x)$), the same R and T apply.

6 Finite Square Well

Potential:

$$V(x) = \begin{cases} -V_0, & -a < x < a, \\ 0, & |x| > a. \end{cases}$$

Here $V_0 > 0$.

6.1 Bound States ($-V_0 < E < 0$)

Define:

$$\kappa = \frac{\sqrt{-2mE}}{\hbar}, \quad l = \frac{\sqrt{2m(E + V_0)}}{\hbar}.$$

For even parity solutions:

$$\psi(x) = \begin{cases} Fe^{-\kappa x}, & x > a, \\ D \cos(lx), & 0 < x < a, \\ \psi(-x), & x < 0. \end{cases}$$

Continuity at $x = a$ gives:

$$\kappa = l \tan(la). \quad (26)$$

For odd parity solutions:

$$\psi(x) = \begin{cases} Fe^{-\kappa x}, & x > a, \\ C \sin(lx), & 0 < x < a, \\ -\psi(-x), & x < 0, \end{cases}$$

leading to:

$$\kappa = -l \cot(la). \quad (27)$$

Define $z = la$ and $z_0 = \frac{a}{\hbar} \sqrt{2mV_0}$. Then $\kappa a = \sqrt{z_0^2 - z^2}$, and:

$$\tan z = \sqrt{(z_0/z)^2 - 1} \quad (\text{even}), \quad \cot z = -\sqrt{(z_0/z)^2 - 1} \quad (\text{odd}).$$

Key results:

- There is always at least one bound state (even).
- As $V_0 \rightarrow \infty$, $E_n + V_0 \approx \frac{n^2 \pi^2 \hbar^2}{2m(2a)^2}$ (infinite square well energies).
- As z_0 decreases, the number of bound states decreases.

6.2 Scattering States ($E > 0$)

Define $k = \sqrt{2mE}/\hbar$ and $l = \sqrt{2m(E + V_0)}/\hbar$. The transmission coefficient is

$$T^{-1} = 1 + \frac{V_0^2}{4E(E + V_0)} \sin^2 \left(\frac{2a}{\hbar} \sqrt{2m(E + V_0)} \right). \quad (28)$$

Perfect transmission ($T = 1$) occurs when:

$$\frac{2a}{\hbar} \sqrt{2m(E_n + V_0)} = n\pi, \quad n = 1, 2, 3, \dots$$

These energies match the infinite square well energies.

7 The Scattering Matrix

For a localized potential, define the incoming amplitudes (A, G) and the outgoing amplitudes (B, F):

$$\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx}, & x \ll 0, \\ Fe^{ikx} + Ge^{-ikx}, & x \gg 0. \end{cases}$$

The S -matrix relates them:

$$\begin{pmatrix} B \\ F \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} A \\ G \end{pmatrix}. \quad (29)$$

Reflection and transmission coefficients:

$$R_l = |S_{11}|^2, \quad T_l = |S_{21}|^2, \quad R_r = |S_{22}|^2, \quad T_r = |S_{12}|^2.$$

Bound states correspond to poles of the S -matrix at $k \rightarrow i\kappa$.

8 Key Equations Summary

Time-independent Schrödinger: $-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V\psi = E\psi.$ (30)

General solution: $\Psi(x, t) = \sum_n c_n \psi_n(x) e^{-iE_n t/\hbar}.$ (31)

Infinite square well: $E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}, \quad \psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right).$ (32)

Harmonic oscillator: $E_n = \left(n + \frac{1}{2}\right) \hbar\omega, \quad \psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n(\xi) e^{-\xi^2/2}.$ (33)

Delta well bound state: $E = -\frac{m\alpha^2}{2\hbar^2}, \quad \psi(x) = \frac{\sqrt{m\alpha}}{\hbar} e^{-m\alpha|x|/\hbar^2}.$ (34)

Finite square well transmission: $T^{-1} = 1 + \frac{V_0^2}{4E(E + V_0)} \sin^2\left(\frac{2a}{\hbar} \sqrt{2m(E + V_0)}\right).$ (35)